

Project Presentation of CO-OP Project at Industry (Module-2) (22CS421)

On

Impact of AI on Education in Young Adults (15-25 Years)

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Artificial Intelligence (AI) is a branch of computer science that focuses on creating systems and machines capable of performing tasks that normally require human intelligence.. In recent years, AI has rapidly developed and become a part of many fields, especially education, where it is transforming the way students learn and teachers teach.

In the education sector, AI is widely used among young adults aged 15–25 years, who are mostly school and college students. AI-powered tools like virtual tutors, smart learning platforms, and language assistants help students understand concepts more easily and complete academic tasks efficiently

Today, AI is not just an additional tool but has become an important support system in education. It helps in research, assignments, coding, writing improvement, and exam preparation. However, its growing use also raises concerns about over-dependence and reduced self-learning abilities, making it important to use AI in a balanced and responsible way.

Problem Statement



- **Over-dependence on AI tools**
Many young students (15–25 years) rely too much on AI for completing assignments and solving academic tasks. This reduces their habit of self-learning and understanding concepts deeply.
- **Decline in critical thinking and originality**
Since AI provides instant answers, students often avoid thinking independently. This leads to weaker problem-solving skills and less creativity in their academic work.
- **Academic misuse and unequal access**
Some students misuse AI for copying assignments (plagiarism), while others do not have equal access to advanced AI tools, creating inequality in learning opportunities.



Positive Impact of AI



- Artificial Intelligence has brought many positive changes in the education system, especially for young adults aged 15–25 years. One of the biggest advantages is personalized learning, where AI understands the learning speed and ability of each student and provides customized study materials accordingly. This helps students learn in a better and more effective way.
- AI also provides quick access to information and explanations, which saves a lot of time during studies and revision. Students can instantly get summaries, solutions, and doubts cleared without waiting for teachers or searching through long textbooks.
- Another important benefit is improved productivity and skill development. AI tools help students in writing assignments, coding practice, exam preparation, and even learning new skills.



Dataset Description

- 100 student records
- Features:
 - AI Usage Hours
 - Study Hours
 - Attendance
 - Previous Marks
 - AI Dependency Index
- Output: Current Marks

Key Learnings

1. AI is transforming education by making learning more personalized, fast, and accessible for students.
2. It acts as a support tool, helping in studying, coding, writing, and understanding complex topics easily.
3. While AI improves efficiency, it is important to use it responsibly to avoid over-dependence and loss of critical thinking skills.
4. The best learning outcome is achieved when AI is combined with traditional classroom learning.
5. Proper awareness and guidance are needed to ensure ethical and balanced use of AI in education.

Methodology Used

- The study follows a predictive and quantitative research design to analyze the impact of AI on education.
- A dataset of 100 student profiles was created with features like AI usage, study hours, attendance, previous marks, and AI dependency index.
- Data preprocessing was performed, including feature selection and calculation of the AI Dependency Index.
- A Supervised Machine Learning model (Linear Regression) was used to predict student academic performance.
- The dataset was split into 80% training data and 20% testing data for model evaluation.
- Model performance was analyzed using graphical visualization and comparison of actual vs predicted results.

- ✓ **AI is transforming education**

It is making learning faster, easier, and more interactive for students.

- ✓ **Helps in better understanding**

AI gives quick explanations and personalized learning support to students.

Improves productivity and skills

It helps in completing assignments, coding, and developing new skills efficiently.

- ✓ **Risk of overuse**

Excess use of AI can reduce creativity, thinking ability, and originality.

Balanced use is important

Combining AI with traditional learning gives the best educational results..

Future Scope



- **Smart Classrooms with AI Integration**
Future classrooms will use AI-powered smart boards, sensors, and interactive systems to make learning more engaging and personalized for students.
- **Virtual Reality (VR) and Augmented Reality (AR) Learning**
Students will learn through immersive experiences like virtual labs, 3D models, and simulations, making complex topics easier to understand.
- **AI-Powered Personalized Education Systems**
AI will analyze each student's learning speed and style to create customized lessons, improving understanding and academic performance.

Thank You